EMBL-EBI

The Job Dispatcher sequence analysis tools services from EMBL-EBI in 2022

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Abstract

The Job Dispatcher (JD) tools framework provides free access to popular bioinformatics sequence analysis applications and sequence libraries. The services can be accessed via user-friendly web interfaces and via established RESTful and SOAP Web Services Application Programming Interfaces (APIs), allowing them to be integrated into third-party systems. The services reported here are integrated into many popular data resources provided at the EMBL-EBI, including for example UniProtKB, ENA, Ensembl Genomes and InterPro. Tools running under JD comprise popular bioinformatics applications (e.g. Clustal Omega, InterProScan, FASTA, NCBI BLAST+, and many more), which combined with nucleotide and protein sequence databases (e.g. UniProtKB and ENA) represent powerful tools for SARS-CoV-2 and COVID-19 research. Countless jobs have been performed on EMBL-EBI's high-performance computing clusters, with a noticeable surge of more than 60 million jobs per month performed during the COVID-19 outbreak months of April and May 2020. Nearly 500 million analyses were executed during 2021 by over 700,000 unique users.

Bioinformatics Tools

Job Dispatcher provides free access to core bioinformatics applications, across a variety of tool categories, including:

Sequence Similarity Search
- NCBI BLAST+
- PSI-Blast
- FASTA
- FASTM/S/F
- SSEARCH
- GGSEARCH
- GLSEARCH
- PSI-Search
- HMMER3 hmmscan
- HMMER3 phmmer

Protein Family Analysis and RNA
- InterProScan 5
- PfamScan
- Infernal cmscan
- Phobius
- Pratt
- Radar
- R2DT

Pairwise Sequence Alignment
- Needle
- Stretcher
- Water
- Matcher
- Lalign
- GeneWise
- GGSEARCH2SEQ
- SSEARCH2SEQ

Multiple Sequence Alignment
- Clustal Omega
- MAFFT
- MUSCLE
- T-coffee
- Kalign
- Prank
- MView
- EMBOSS cons

Sequence Operations and Statistics
- EMBOSS seqret
- EMBOSS transeq
- EMBOSS sixpack
- EMBOSS backtranseq
- EMBOSS backtranambig
- EMBOSS pepinfo
- EMBOSS pepstats
- EMBOSS pepwindow
- EMBOSS cpgplot
- EMBOSS newcpgreport
- EMBOSS isochore
- EMBOSS dotmatcher
- EMBOSS dotpath
- EMBOSS dottup
- EMBOSS polydot
- EMBOSS cons
- Seqchksum
- Simple Phylogeny

Job Dispatcher services can be accessed via webforms or via SOAP and REST APIs and are available through ELIXIR's bio.tools. Web-service example clients in Python, Perl and Java, as well as Common Workflow Language (CWL) example workflows are also provided.

Recent Updates and Ongoing Work

- Added new tools: R2DT, SSEARCH2SEQ and GGSEARCH2SEQ
- Added new datasets: AlphaFold DB (latest v3 release contain 214,684,312 entries), UniVec, EMDB and PDBe-KB (the latter two available on Dbfetch)
- Launched new blog at https://ebi-jdispatcher.github.io/blog/
- Currently working on a new full-featured and interactive frontend developed with Angular framework
- The next iteration of Job Dispatcher (JD2) which is based on Java Spring Boot and features a refreshed RESTful API, is under active development

Service Usage Highlights

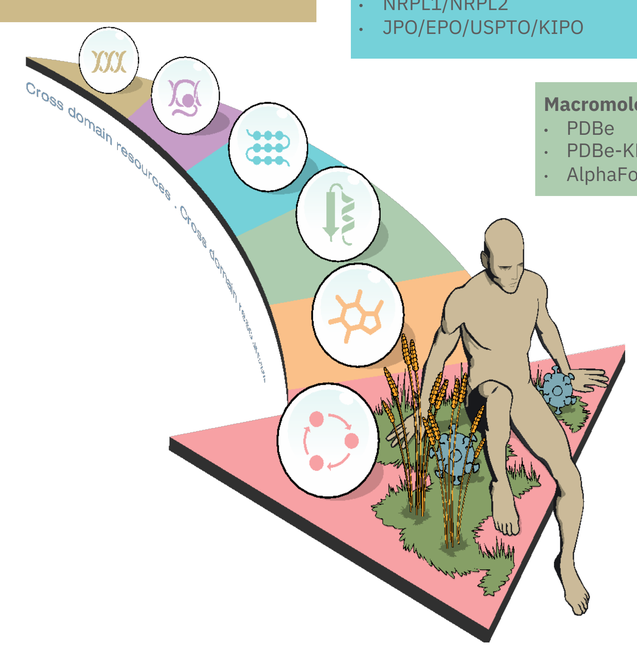
- ~130 million jobs during the COVID-19 outbreak months, April and May 2020
- ~500 million jobs in 2021, and ~300 million jobs so far in 2022

Biological Data Retrieval

Biological data fetching with Dbfetch is available through web and a common REST API. Our data resources are delivered in partnership with collaborators, including: UniProt, ENA, Ensembl, WormBase, InterPro, PDBe, IPD/IMGT, EBI Search and among others.

Nucleotide Sequences
- Ensembl Genomes
- ENA releases and updates
- ENA coding
- ENA non-coding
- ENA geospatial
- ENA barcode
- NRNL1/NRNL2
- EMVec
- UniVec
- Emsembl 2020

Protein Sequences
- UniProtKB Swissprot
- UniProtKB TrEMBL
- UniProtKB Taxonomy Subsets
- UniProtKB Reference Proteomes
- UniRef
- UniParc
- NRPL1/NRPL2
- JPO/EPO/USPTO/KIPO



Macromolecular Structures
- PDBe
- PDBe-KB
- AlphaFold DB

Specialized Databases
- IPD-IMGT HLA/MHC/KIR/NHKIR
- Vectors
- ChEMBL
- MEROPS
- IntAct
- Enzyme Portal

Other Datasets
- NCBI Taxonomy
- EMDB
- MEDLINE
- EDAM Ontology
- HGNC

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References

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